# Volumes of revolution 5

## Objectives

After completing this chapter you should be able to:

- Find the volume of revolution when a curve is rotated around the $x$-axis → pages 72–75
- Find the volume of revolution when a curve is rotated around the $y$-axis → pages 76–78
- Find more complicated volumes of revolution → pages 78–83
- Model real-life objects using volumes of revolution → pages 83–86

## Prior knowledge check

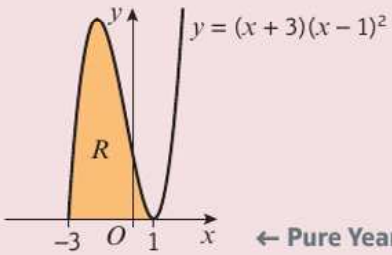
**1** Evaluate:

**a** $\int_2^4 (6x^2 - 8x)\,\mathrm{d}x$

**b** $\int_4^9 \left(\frac{5}{\sqrt{x}} + x^{\frac{3}{2}}\right)\mathrm{d}x$

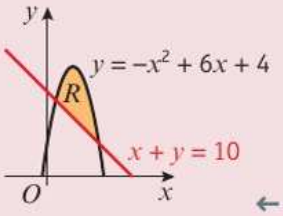
**c** $\int_1^2 \frac{x^2 + 8x}{x^4}\,\mathrm{d}x$ ← Pure Year 1, Chapter 13

**2** Find the area of the region $R$ bounded by the curve $y = (x + 3)(x - 1)^2$ and the $x$-axis.




← Pure Year 1, Chapter 13

**3** Find the area of the finite region bounded by the curve $y = -x^2 + 6x + 4$ and the line $x + y = 10$.



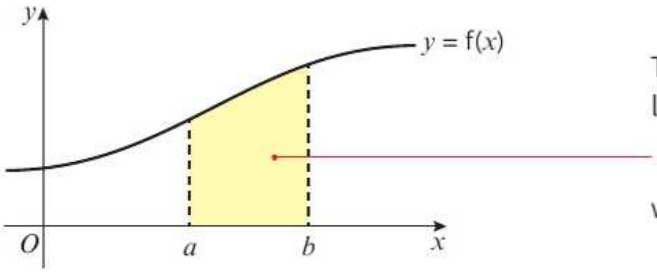

← Pure Year 1, Chapter 13

Woodworkers use lathes to create solid objects that have circular cross-sections. Solids such as this are called volumes of revolution, and you can find their volumes using calculus.

## 5.1 Volumes of revolution around the $x$-axis

You have used integration to find the area of a region $R$ bounded by a curve, the $x$-axis and two vertical lines.

**Notation** This process is called **definite integration**. ← Pure Year 1, Sections 13.4, 13.5




The area between a positive curve, the $x$-axis and the lines $x = a$ and $x = b$ is given by

$$\text{Area} = \int_a^b y\,\mathrm{d}x$$

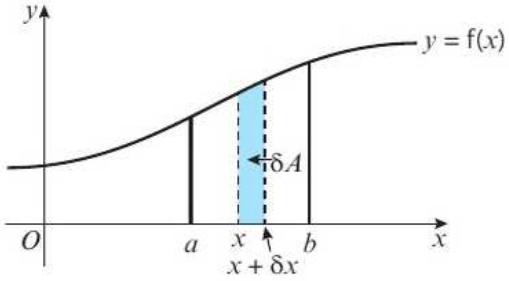
where $y = \mathrm{f}(x)$ is the equation of the curve.

You can derive this formula by considering the sum of an infinite number of small strips of width $\delta x$. Each of these strips has a height of $y$, so the area of each strip is

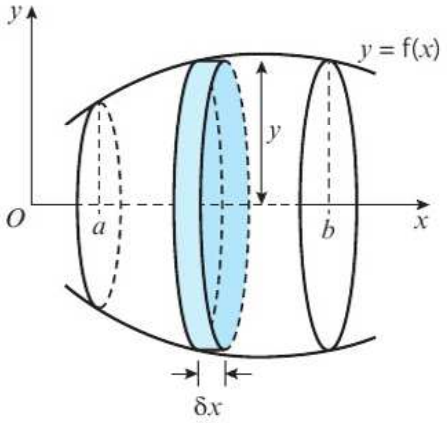
$$\delta A = y\delta x$$

The total area is approximately the sum of these strips, or $\sum y\delta x$.

The exact area is the limit of this sum as $\delta x \to 0$, which is written as $\int y\,\mathrm{d}x$.




You can use a similar technique to find the volume of an object created by rotating a curve around a coordinate axis. If each of these strips is rotated through $2\pi$ radians (or 360°) about the $x$-axis, it will form a shape that is approximately cylindrical. The volume of each cylinder will be $\pi y^2\delta x$ since it will have radius $y$ and height $\delta x$.



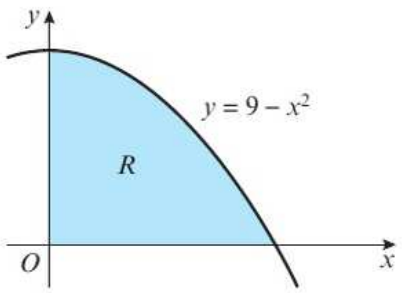

So the volume of the solid will be approximately equal to the sum of the volumes of each cylinder, or $\sum \pi y^2\delta x$. The exact volume is the limit of this sum as $\delta x \to 0$, or $\pi\int y^2\,\mathrm{d}x$.

- **The volume of revolution formed when $y = \mathrm{f}(x)$ is rotated through $2\pi$ radians about the $x$-axis between $x = a$ and $x = b$ is given by:**

$$\textbf{Volume} = \pi\int_a^b y^2\,\mathrm{d}x$$

## Example 1

The diagram shows the region $R$ which is bounded by the $x$-axis, the $y$-axis and the curve with equation $y = 9 - x^2$. The region is rotated through $2\pi$ radians about the $x$-axis. Find the exact volume of the solid generated.




| | |
|---|---|
| $9 - x^2 = 0$ | First find the point where the curve intersects the $x$-axis. |
| $(3 - x)(3 + x) = 0$ | |
| $x = 3$ or $x = -3$ | From the diagram, $x > 0$, therefore $x = 3$. |
| $V = \pi\int_0^3 (9 - x^2)^2\,dx$ | Use $V = \pi\int_a^b y^2\,dx$ with $a = 0$, $b = 3$ and $y = 9 - x^2$. |
| $= \pi\int_0^3 (81 - 18x^2 + x^4)\,dx$ | Simplify the integrand. |
| $= \pi\left[81x - 6x^3 + \frac{1}{5}x^5\right]_0^3$ | Integrate each term separately. |
| $= \pi\left(\left(243 - 162 + \frac{243}{5}\right) - (0 - 0 + 0)\right)$ | Substitute the limits. |
| $= \frac{648\pi}{5}$ | Simplify the resulting answer and write it as an exact fraction in terms of $\pi$. |

## Exercise 5A

**1** Find the exact volume of the solid generated when each curve is rotated through 360° about the $x$-axis between the given limits.

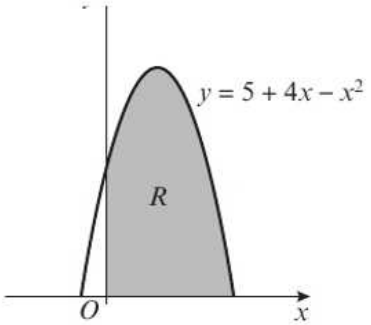
**a** $y = 10x^2$ between $x = 0$ and $x = 2$

**b** $y = 5 - x$ between $x = 3$ and $x = 5$

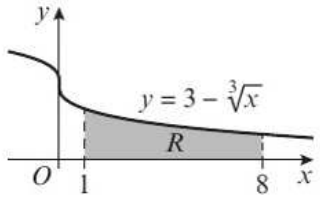
**c** $y = \sqrt{x}$ between $x = 2$ and $x = 10$

**d** $y = 1 + \frac{1}{x^2}$ between $x = 1$ and $x = 2$

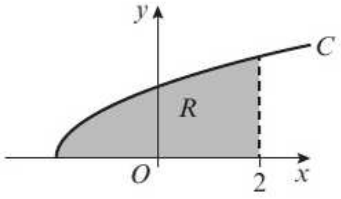
E **2** The curve shown in the diagram has equation $y = 5 + 4x - x^2$. The finite region $R$ is bounded by the curve, the $x$-axis and the $y$-axis. The region is rotated through $2\pi$ radians about the $x$-axis to generate a solid of revolution. Find the exact volume of the solid generated. **(5 marks)**



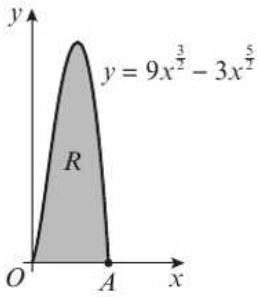
E **3** The diagram shows the region $R$ which is bounded by the $x$-axis, the lines $x = 1$ and $x = 8$, and the curve with equation $y = 3 - \sqrt[3]{x}$. The region is rotated through $2\pi$ radians about the $x$-axis. Find the exact volume of the solid generated. **(5 marks)**




E **4** The diagram shows the curve $C$ with equation $y = \sqrt{x + 2}$. The region $R$ is bounded by the $x$-axis, the line $x = 2$ and $C$. The region is rotated through 360° about the $x$-axis. Find the exact volume of the solid generated. **(5 marks)**




E/P **5** The diagram shows a sketch of the curve with equation $y = 9x^{\frac{3}{2}} - 3x^{\frac{5}{2}}$. The region $R$ is bounded by the curve and the $x$-axis.

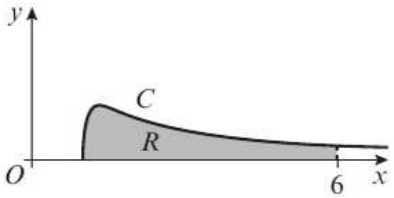



**a** Find the coordinates of $A$. **(2 marks)**

The region is rotated through $2\pi$ radians about the $x$-axis.

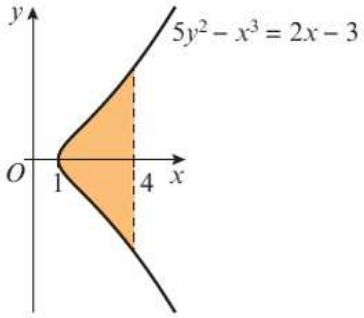
**b** Find the volume of the solid of revolution generated. **(5 marks)**

E/P **6** The curve with equation $y = \dfrac{\sqrt{3x^4 - 3}}{x^3}$ is shown in the diagram.




The region bounded by the curve $C$, the $x$-axis and the line $x = 6$ is shown shaded in the diagram. The region is rotated through $2\pi$ radians about the $x$-axis. Find the volume of the solid generated, giving your answer correct to 3 significant figures. **(6 marks)**

(P) **7** The diagram shows the curve with equation $5y^2 - x^3 = 2x - 3$. The shaded region is bounded by the curve and the line $x = 4$. The region is rotated about the $x$-axis to generate a solid of revolution. Find the volume of the solid generated.

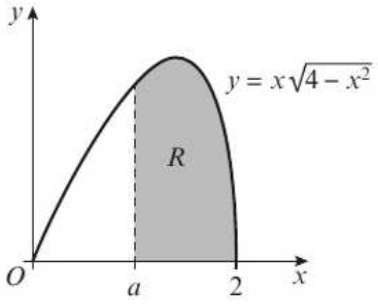



**Problem-solving**

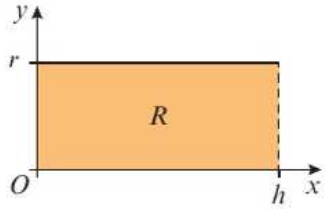
Rearrange the equation to make $y^2$ the subject.

(E/P) **8** The curve shown in the diagram has equation $y = x\sqrt{4 - x^2}$. The finite region $R$ is bounded by the curve, the $x$-axis and the line $x = a$, where $0 < a < 2$. The region is rotated through $2\pi$ radians about the $x$-axis to generate a solid of revolution with volume $\frac{657\pi}{160}$

Find the value of $a$. **(5 marks)**

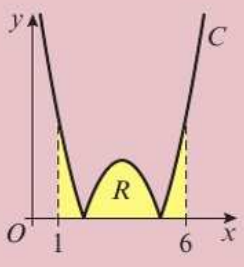



(P) **9** The diagram shows a shaded rectangular region $R$ of length $h$ and width $r$. The region $R$ is rotated through 360° about the $x$-axis. Use integration to show that the volume, $V$, of the cylinder formed is $V = \pi r^2 h$.



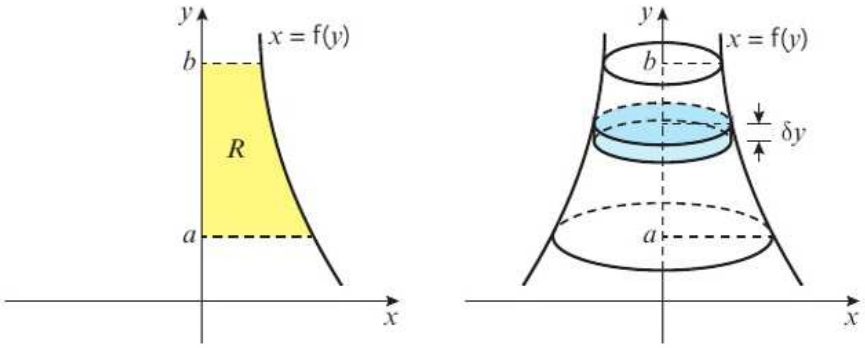

**Challenge**

The diagram shows the curve $C$ with equation $y = |x^2 - 7x + 10|$. The shaded region $R$ is bounded by the $x$-axis, the curve $C$ and the lines $x = 1$ and $x = 6$. The region is rotated $2\pi$ radians about the $x$-axis. Find the exact volume of the solid generated.



## 5.2 Volumes of revolution around the $y$-axis

You can find a volume of revolution around the $y$-axis by considering $x$ as a function of $y$. The diagram shows a curve with equation $x = f(y)$. A small strip of height $\delta y$ is rotated $2\pi$ radians about the $y$-axis. The volume of the cylinder created will be $\pi x^2 \delta y$ since the radius is $x$ and the height is $\delta y$.




So when the whole region, $R$, is rotated $2\pi$ radians about the $y$-axis, the volume formed will be approximately equal to the sum of the volumes of each cylinder, or $\sum \pi x^2 \delta y$. The exact volume is the limit of this sum as $\delta x \to 0$, or $\pi \int x^2 \, dy$.

- **The volume of revolution formed when $x = f(y)$ is rotated through $2\pi$ radians about the $y$-axis between $y = a$ and $y = b$ is given by**

$$\textbf{Volume} = \pi \int_a^b x^2 \, dy$$

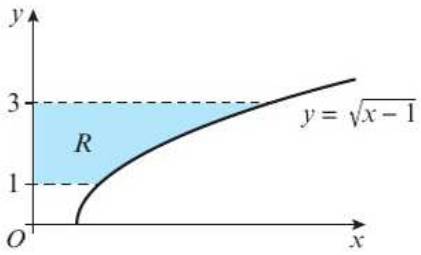
**Online** Explore volumes of revolution around the $x$- and $y$-axes using GeoGebra.



### Example 2

The diagram shows the curve with equation $y = \sqrt{x - 1}$. The region $R$ is bounded by the curve, the $y$-axis and the lines $y = 1$ and $y = 3$.

The region is rotated through 360° about the $y$-axis. Find the volume of the solid generated.




$y = \sqrt{x - 1}$
$y^2 = x - 1$
$x = y^2 + 1$

First rearrange the equation to make $x$ the subject.

$V = \pi \int_1^3 (y^2 + 1)^2 dy$

Use $V = \pi \int_a^b x^2 \, dy$ with $a = 1$, $b = 3$ and $x = y^2 + 1$.

$= \pi \int_1^3 (y^4 + 2y^2 + 1) \, dy$

Simplify the integrand.

$= \pi \left[\frac{1}{5}y^5 + \frac{2}{3}y^3 + y\right]_1^3$

Integrate each term separately.

$= \pi\left(\left(\frac{243}{5} + 18 + 3\right) - \left(\frac{1}{5} + \frac{2}{3} + 1\right)\right)$

Substitute the limits.

$= \pi\left(\frac{1044}{15} - \frac{28}{15}\right)$

$= \frac{1016\pi}{15}$

Simplify the resulting answer and write it as an exact fraction in terms of $\pi$.

## Exercise 5B

**1** Find the exact volume of the solid generated when each curve is rotated through 360° about the $y$-axis between the given limits.

**Hint** In part **d**, rearrange the expression to make $x^2$ the subject.

**a** $x = \frac{1}{2}y + 1$ between $y = 2$ and $y = 5$

**b** $y = 2\sqrt{x}$ between $y = 0$ and $y = 1$

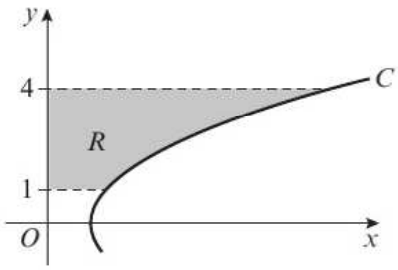
**c** $y = \frac{1}{x}$ between $y = 1$ and $y = 3$

**d** $y = 2x^2 - 4$ between $x = 5$ and $x = 11$

(E) **2** The curve $C$ with equation $x = \frac{1}{2}y^2 + 1$ is shown in the diagram.

The region $R$ is bounded by the lines $y = 1$, $y = 4$, the $y$-axis and the curve $C$, as shown in the diagram. The region is rotated through $2\pi$ radians about the $y$-axis.

Find the volume of the solid generated. **(6 marks)**

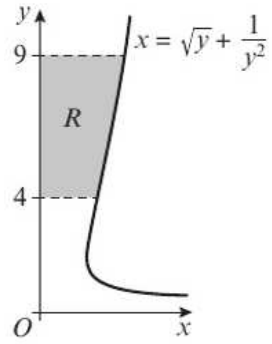



(E) **3** The diagram shows the finite region $R$, which is bounded by the curve $x = \sqrt{y} + \frac{1}{y^2}$, the lines $y = 4$, $y = 9$ and the $y$-axis.

**a** Find the exact area of the shaded region. **(3 marks)**

The region $R$ is rotated through $2\pi$ radians about the $y$-axis.

**b** Use integration to find the volume of the solid generated. Round your answer to 2 decimal places. **(5 marks)**

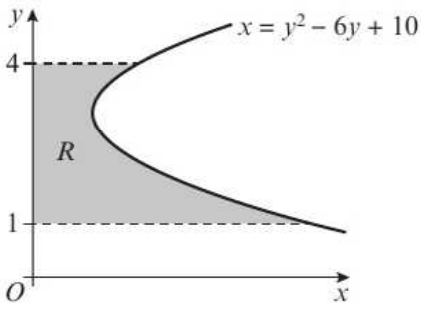



(E) **4** The diagram shows the finite region $R$, which is bounded by the curve $x = y^2 - 6y + 10$, the lines $y = 1$, $y = 4$ and the $y$-axis.

**a** Find the area of the shaded region $R$. **(3 marks)**

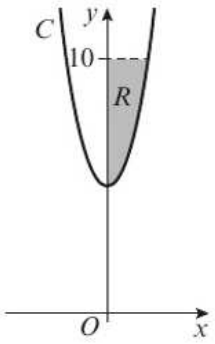
The region $R$ is rotated through 360° about the $y$-axis.

**b** Use integration to find an exact value for the volume of the solid generated. **(5 marks)**



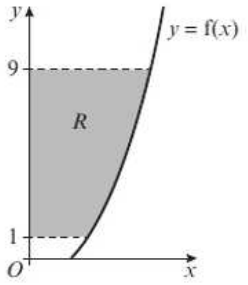

(E/P) **5** The curve $C$ with equation $y = 2x^2 + 5$ is shown in the diagram.

The region bounded by the $y$-axis, the curve $C$ and the line $y = 10$ is shown and shaded in the diagram. The region is rotated 360° about the $y$-axis. Find the exact volume of the solid generated. **(6 marks)**



E **6** $f(x) = x^2 - 2x + 1, x \geqslant 1$

The diagram shows the finite region $R$ bounded by the curve $y = f(x)$, the $y$-axis and the lines $y = 1$ and $y = 9$.



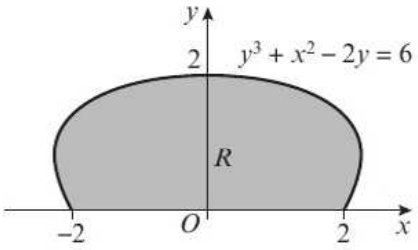

**a** Show that the equation $y = f(x)$ can be written as $x^2 = y + 2\sqrt{y} + 1$. **(2 marks)**

**b** The region $R$ is rotated through $2\pi$ radians about the $y$-axis. Find the exact volume of the solid generated. **(5 marks)**

E/P **7** The diagram shows the finite region $R$, which is bounded by the curve $y^3 + x^2 - 2y = 4$ and the $x$-axis.

The region $R$ is rotated about the $y$-axis to generate a solid of revolution.

Find an exact value for the volume of the solid. **(5 marks)**

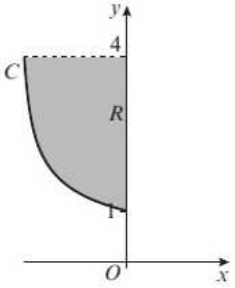



E/P **8** Part of the curve $C$ with equation $y^2 = \dfrac{1}{2x + 1}$ is shown in the diagram.

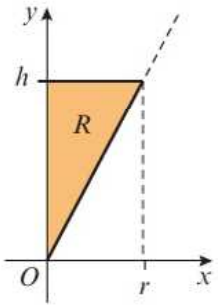
The region $R$ is bounded by the curve, the $y$-axis and the line $y = 4$.

The region $R$ is rotated $2\pi$ radians about the $y$-axis.

Find the volume of the solid generated. **(5 marks)**




P **9** The diagram shows a shaded region $R$ in the shape of a right-angled triangle of width $r$ and height $h$. The region $R$ is rotated through $2\pi$ radians about the $y$-axis. Use integration to show that the volume, $V$, of the cone formed is given by $V = \frac{1}{3}\pi r^2 h$.




**Problem-solving**

Start by finding an equation for the line that forms the hypotenuse of the triangle.

## 5.3 Adding and subtracting volumes

You might need to solve volume of revolution problems involving the volumes of cylinders and cones. Remember these two formulae:

- **A cylinder of height $h$ and radius $r$ has volume $\pi r^2 h$.**
- **A cone of height $h$ and base radius $r$ has volume $\frac{1}{3}\pi r^2 h$.**

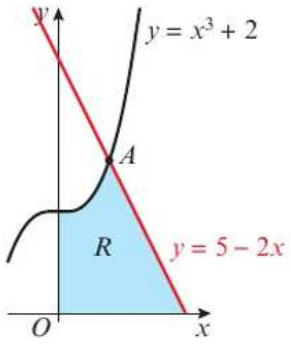
## Example 3

The region $R$ is bounded by the curve with equation $y = x^3 + 2$, the line $y = 5 - 2x$, and the $x$- and $y$-axes.

**a** Verify that the coordinates of $A$ are (1, 3).

A solid is created by rotating the region 360° about the $x$-axis.

**b** Find the volume of this solid.



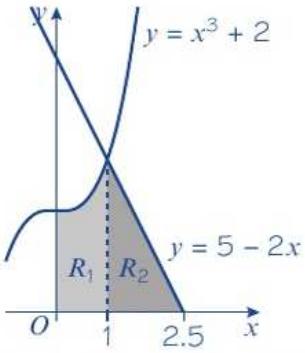

**a** Curve: $1^3 + 2 = 3$

Line: $5 - 2 \times 1 = 3$

So (1, 3) is the point of intersection.

Substitute $x = 1$ into each equation to show that the point (1, 3) lies on both the line and the curve.

**b** Find the volumes of revolution of $R_1$ and $R_2$ separately.




**Problem-solving**

Divide the original area into two separate areas. Use integration to find the volume of revolution of $R_1$. Then add the volume of the cone formed by rotating $R_2$ about the $x$-axis.

### Volume of revolution of $R_1$

$V = \pi\int_0^1 (x^3 + 2)^2\,dx$

Use $V = \pi\int_a^b y^2\,dx$ with $a = 0$, $b = 1$ and $y = x^3 + 2$.

$= \pi\int_0^1 (x^6 + 4x^3 + 4)\,dx$

Simplify the integrand.

$= \pi\left[\frac{1}{7}x^7 + x^4 + 4x\right]_0^1$

Integrate each term separately.

$= \pi\left(\left(\frac{1}{7} + 1 + 4\right) - (0 + 0 + 0)\right)$

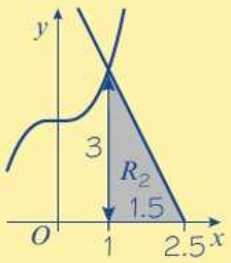
Substitute the limits.

$= \frac{36\pi}{7}$

Simplify the resulting answer by writing it as an exact fraction in terms of $\pi$.

### Volume of revolution of $R_2$

The line $y = 5 - 2x$ will intersect the $x$-axis at $x = 2.5$.




When $R_2$ is rotated about the $x$-axis, it will create a cone.

The height of the cone is $2.5 - 1 = 1.5$.

The radius of the cone is 3.

$V = \frac{1}{3}\pi \times 3^2 \times 1.5$

The formula for the volume of a cone is $V = \frac{1}{3}\pi r^2 h$.

$V = \frac{9\pi}{2}$

The total volume is $\frac{36\pi}{7} + \frac{9\pi}{2} = \frac{135\pi}{14}$

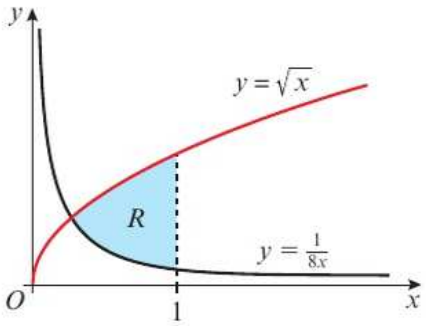
Add the values of $R_1$ and $R_2$ to find the total volume of revolution.

You can find more complicated volumes of revolution by subtracting one volume of revolution from another.

## Example 4

The diagram shows the region $R$ bounded by the curves with equations $y = \sqrt{x}$ and $y = \frac{1}{8x}$ and the line $x = 1$.

The region is rotated through 360° about the $x$-axis. Find the exact volume of the solid generated.




At point of intersection,

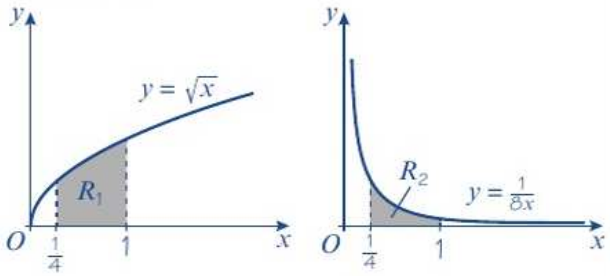
$\sqrt{x} = \frac{1}{8x}$

$x^{\frac{3}{2}} = \frac{1}{8}$

$x = \left(\frac{1}{8}\right)^{\frac{2}{3}} = \frac{1}{4}$

Solve the equations simultaneously to find the $x$-coordinate at the point of intersection of the two curves.

Consider volumes of revolution of $R_1$ and $R_2$ separately:




**Volume of revolution of $R_1$:**

$V_1 = \pi \int_{\frac{1}{4}}^{1} (\sqrt{x})^2 \, dx$

$= \pi \left[\frac{1}{2}x^2\right]_{\frac{1}{4}}^{1}$

$= \pi\left(\frac{1}{2} - \frac{1}{32}\right) = \frac{15\pi}{32}$

**Volume of revolution of $R_2$:**

$V_2 = \pi \int_{\frac{1}{4}}^{1} \left(\frac{1}{8x}\right)^2 dx$

$= \pi \int_{\frac{1}{4}}^{1} \frac{1}{64x^2} dx$

$= \pi\left[-\frac{1}{64x}\right]_{\frac{1}{4}}^{1}$

$= \pi\left(-\frac{1}{64} - \left(-\frac{1}{16}\right)\right) = \frac{3\pi}{64}$

**Volume of revolution of $R$:**

$V = \frac{15\pi}{32} - \frac{3\pi}{64} = \frac{27\pi}{64}$

**Problem-solving**

Work out how you can create the necessary volume of revolution using simpler volumes of revolution. In this case, you can find the volume of revolution of the area under the $y = \sqrt{x}$ curve, then subtract the volume of revolution of the area under the $y = \frac{1}{8x}$ curve.

$\frac{1}{64x^2} = \frac{1}{64}x^{-2}$, so when you integrate this term becomes $\frac{\frac{1}{64}x^{-1}}{-1} = -\frac{1}{64x}$

The safest way to find this volume is to work out each volume of revolution separately then subtract, as shown here. But you could also do this in one combined calculation:

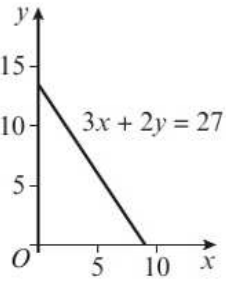
$V = \pi \int_{\frac{1}{4}}^{1} (\sqrt{x})^2 \, dx - \pi \int_{\frac{1}{4}}^{1} \left(\frac{1}{8x}\right)^2 dx$

$= \pi \int_{\frac{1}{4}}^{1} \left((\sqrt{x})^2 - \left(\frac{1}{8x}\right)^2\right) dx = \pi \left[\frac{1}{2}x^2 + \frac{1}{64x}\right]_{\frac{1}{4}}^{1}$

$= \pi\left(\left(\frac{1}{2} + \frac{1}{64}\right) - \left(\frac{1}{32} + \frac{1}{16}\right)\right) = \frac{27\pi}{64}$

## Exercise 5C

**1** The diagram shows the line with equation $3x + 2y = 27$.

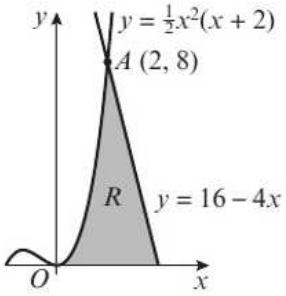



**a** Use integration to find the volume of revolution when the region is rotated through 360° about the $x$-axis.

**b** Use integration to find the volume of revolution when the region is rotated through 360° about the $y$-axis.

**c** Use the formula for the cone to check your answers to parts **a** and **b**. Clearly state the radius and the height in each case.

E/P **2** The region $R$ is bounded by the curve with equation $y = \frac{1}{2}x^2(x + 2)$, the line $y = 16 - 4x$, and the $x$-axis.

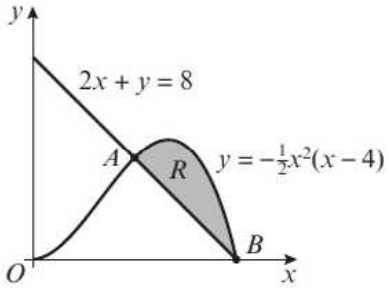



**a** Show that the coordinates of $A$ are (2, 8). **(1 mark)**

A solid is created by rotating the region through 360° about the $x$-axis.

**b** Find the volume of this solid. **(6 marks)**

E/P **3** The region $R$ is bounded by the curve with equation $y = -\frac{1}{2}x^2(x - 4)$ and the line $2x + y = 8$.




**Problem-solving**

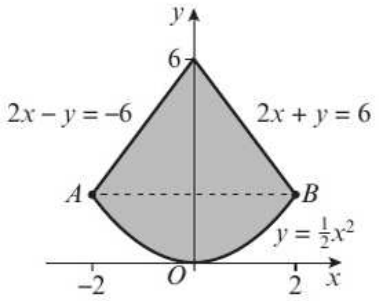
You will need to find a volume of revolution then **subtract** the volume of the cone.

**a** Show that the coordinates of $A$ are (2, 4) and write down the coordinates of $B$. **(1 mark)**

A solid is created by rotating the region through 360° about the $x$-axis.

**b** Find the volume of this solid. **(6 marks)**

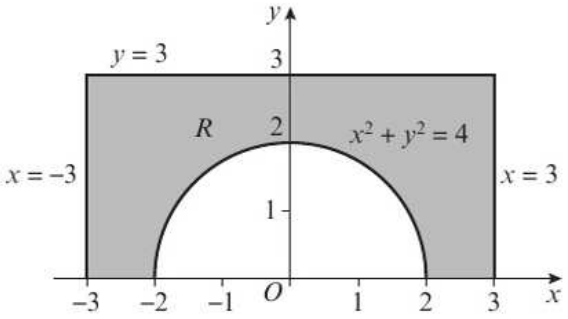
E/P **4** The shape shown is bounded by the curve $y = \frac{1}{2}x^2$, and the lines $2x + y = 6$ and $2x - y = -6$.




**a** Find the coordinates of the points $A$ and $B$. **(2 marks)**

**b** The shape is rotated about the $y$-axis to generate a solid of revolution.
Find the volume of the solid generated. **(6 marks)**

E/P **5** The region $R$ is bounded by the lines $x = -3$, $x = 3$ and $y = 3$, the curve $C$ with equation $x^2 + y^2 = 4$ and the $x$-axis. The region is rotated about the $y$-axis to generate a solid of revolution. Find the volume of the solid generated.

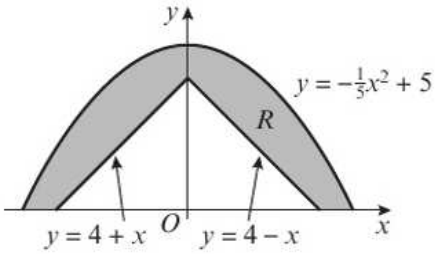



**Problem-solving**

Find the volume generated by rotating the curve $x^2 + y^2 = 4$ about the $x$-axis and subtract this from the volume of a suitable cylinder.

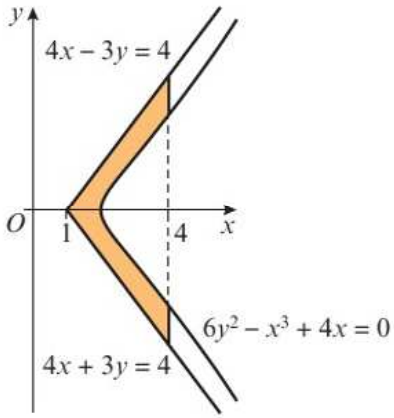
**(6 marks)**

E/P **6** The shaded region $R$ is bounded by the curve $y = -\frac{1}{5}x^2 + 5$, the $x$-axis and the lines with equations $y = 4 - x$ and $y = 4 + x$.



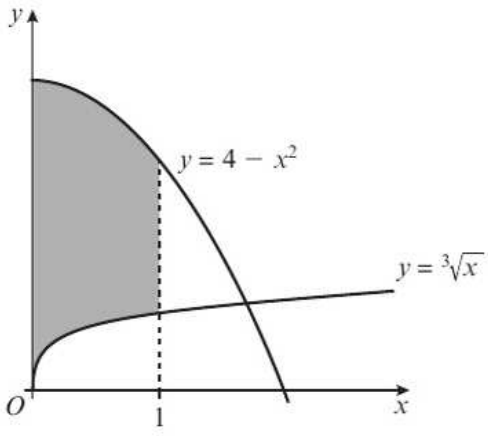

Find the volume of the solid of revolution generated when this region is rotated about the $y$-axis. **(8 marks)**

P **7** The shaded region is bounded by the curve $C$ with equation $6y^2 - x^3 + 4x = 0$, $x > 0$, the straight lines $4x - 3y = 4$, $4x + 3y = 4$, and the line $x = 4$. The region is rotated about the $x$-axis to generate a solid of revolution. Find the exact volume of the solid generated.

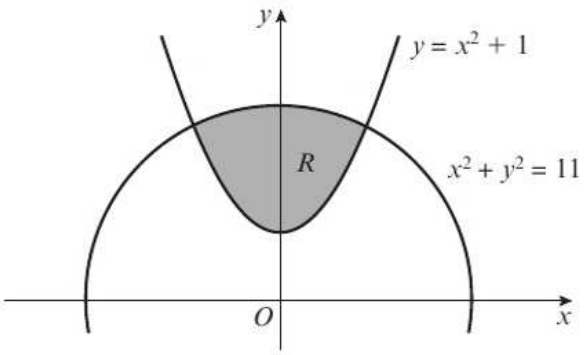



E/P **8** The shaded region is bounded by the curve with equation $y = 4 - x^2$, the curve with equation $y = \sqrt[3]{x}$, the $y$-axis and the line with equation $x = 1$.

The region is rotated through 360° about the $x$-axis. Find the exact volume of the solid generated. **(7 marks)**



E/P **9** The diagram shows the region $R$ bounded by the curve with equation $y = x^2 + 1$ and the curve with equation $x^2 + y^2 = 11$.



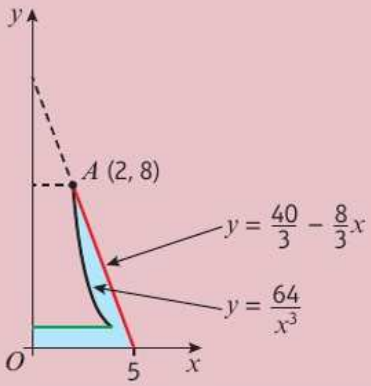

**a** Find the $x$-coordinates of the points of intersection of the two curves. **(3 marks)**

The region $R$ is rotated through 360° about the $x$-axis.

**b** Find the volume of the solid generated, giving your answer correct to 2 decimal places. **(7 marks)**

**Challenge**

The shaded region shown in the diagram is bounded by the curve $y = \frac{64}{x^3}$, the line $y = \frac{40}{3} - \frac{8}{3}x$, the $x$-axis, and the line $y = 1$, for $0 \leqslant x \leqslant 4$. The region is rotated about the $y$-axis. Find an exact value for the volume of the solid generated.

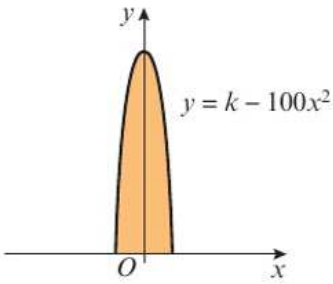



## 5.4 Modelling with volumes of revolution

Volumes of revolution can be used to model real-life situations.

**Example 5**

A manufacturer wants to cast a prototype for a new design for a pen barrel out of solid resin. The shaded region shown in the diagram is used as a model for the cross-section of the pen barrel. The region is bounded by the $x$-axis and the curve with equation $y = k - 100x^2$, and will be rotated around the $y$-axis. Each unit on the coordinate axes represents 1 cm.




**a** Suggest a suitable value for $k$.

**b** Use your value of $k$ to estimate the volume of resin needed to make the prototype.

**c** State one limitation of this model.

**a** $k = 10$.

**b** $y = 10 - 100x^2$

$100x^2 = 10 - y$

$x^2 = \frac{1}{10} - \frac{y}{100}$

$V = \pi \int_0^{10} \left(\frac{1}{10} - \frac{y}{100}\right) dy$

$= \pi \left[\frac{y}{10} - \frac{y^2}{200}\right]_0^{10}$

$= \pi\left(\left(\frac{10}{10} - \frac{10^2}{200}\right) - \left(\frac{0}{10} - \frac{0^2}{200}\right)\right)$

$= \pi\left(1 - \frac{1}{2}\right) = \frac{\pi}{2}$

Approximately 1.57 cm$^3$ of resin will be needed.

**c** The cross-section of the pen is unlikely to match the curve exactly.

Consider the context of the question and choose a value that makes sense. Most pens are between 10 cm and 15 cm long so any value in the range $10 \leqslant k \leqslant 15$ is sensible.

Use your value of $k$ from part **a**.

Rearrange the expression to make $x^2$ the subject.

Use $V = \pi \int_a^b x^2 \, dy$ with $a = 0$, $b = 10$ and $x^2 = \frac{1}{10} - \frac{y}{100}$

Give units with your answer.

**Problem-solving**

You can give any sensible answer that refers to the context of the question. You could also say that some resin might be wasted when the pen is made, or there might be air bubbles in the mould.

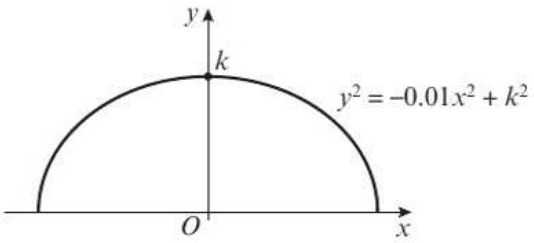
## Exercise 5D

E/P **1** The diagram shows the shape of a large tent at a fair. The outside of the tent can be modelled by the equation $y^2 = -0.01x^2 + k^2$. Each unit on the coordinate axes represents 1 metre.

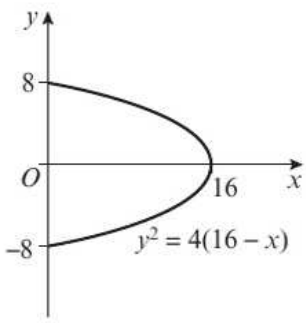
**a** Suggest a suitable value for $k$. **(1 mark)**

**b** Use your value of $k$ to estimate the capacity of the tent. **(5 marks)**

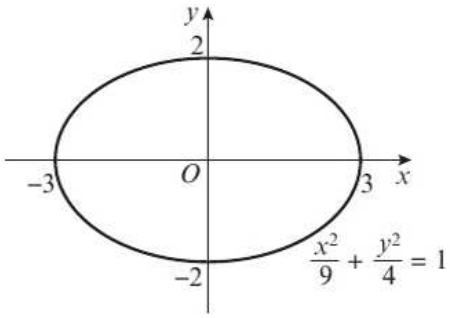
**c** State one limitation of this model. **(1 mark)**




E **2** The diagram shows half of the outline of a rugby ball. The outline is modelled by the curve $y^2 = 4(16 - x)$.
The measurements shown are given in centimetres. By rotating the curve through 360° around the $y$-axis, find the total volume of the rugby ball. **(5 marks)**



**3** The cross-section of an egg can be modelled as an ellipse with equation $\frac{x^2}{9} + \frac{y^2}{4} = 1$, where the dimensions shown are in centimetres.

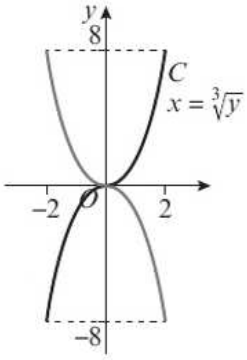



**a** Calculate the volume of the solid formed by rotating this curve through 360° about the $x$-axis.

**b** Show that the solid formed by rotating the curve through 360° about the $y$-axis has the same volume.

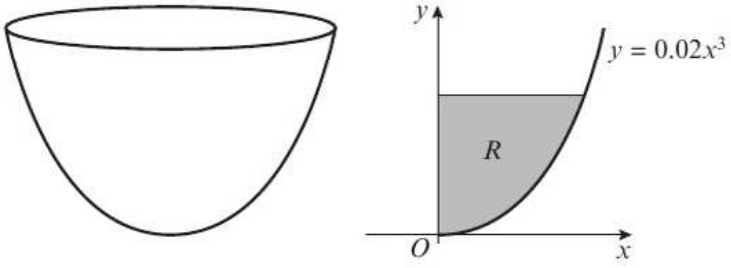
**c** Say which of these two solids most resembles an egg.

E/P **4** The diagram shows the cross-section of an egg timer, which has a height of 16 cm. The shape of the egg timer is modelled as a solid of revolution of a curve $C$ about the $y$-axis. The curve $C$ has equation $x = \sqrt[3]{y}$.




Sand flows through the egg timer at a rate of 8 cm³/min. The designer wants the egg timer to empty in 5 minutes. Calculate, to 2 decimal places, the height of sand that should be placed in the top half of the egg timer. **(5 marks)**

E/P **5** The diagram shows the bowl of an electric stand mixer. The height of the bowl is 18 cm. The shape of the bowl is modelled by rotating the curve with equation $y = 0.02x^3$ through $2\pi$ radians about the $y$-axis.




**a** Find the diameter of the bowl. **(2 marks)**

**b** Find the maximum volume of liquid that can be contained within the mixing bowl. **(4 marks)**

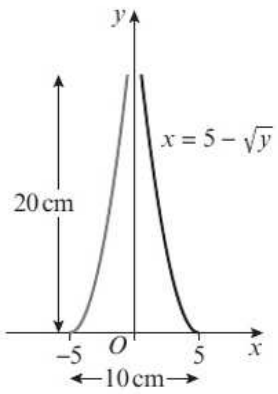
The mixing bowl has a paddle of height 12 cm. The paddle just touches the side of the bowl. In its starting position, the paddle forms a region $R$, as shown in the diagram.

**c** Calculate the area of the paddle. **(3 marks)**

The paddle rotates about the $y$-axis when the mixer is in operation.

**d** Find the proportion of the total volume contained within the bowl that can be mixed by the paddle. **(4 marks)**

E/P **6** The diagram shows a vase with a base width of 10 cm and a height of 20 cm. The edge of the vase is modelled by the equation $x = 5 - \sqrt{y}$. The vase is formed by rotating the shape through 360° about the $y$-axis.

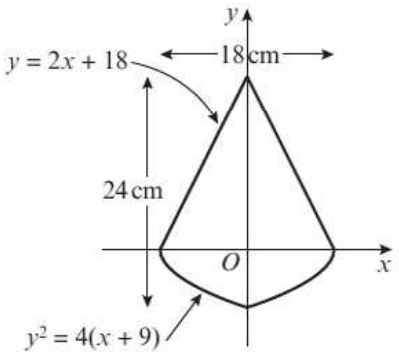



**a** Use this model to estimate the capacity of the vase. **(5 marks)**

The vase is initially filled to a height of 10 cm. When the flowers are placed in the vase, 50 $cm^3$ of water is displaced.

**b** Determine whether the vase will overflow. **(3 marks)**

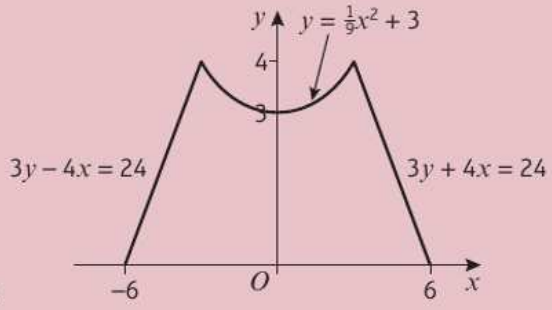
E/P **7** A circular spinning top is made of solid wood with a width of 18 cm and a height of 24 cm. A cross-section of the spinning top is shown in the diagram. The cross-section is formed by part of the curve with equation $y^2 = 4(x + 9)$ and the straight line with equation $y = 2x + 18$, and is symmetrical about the $y$-axis. The cross-section is rotated about the $y$-axis. Find the total volume of wood in the spinning top. **(7 marks)**




## Challenge

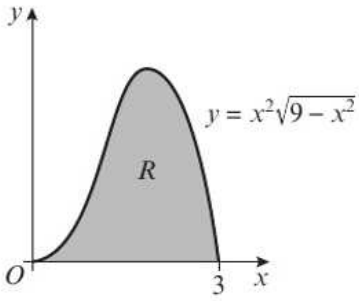
The diagram shows the cross-section of a circular place-holder used to hold a rugby ball when penalties and conversions are kicked. The cross-section can be modelled by the straight lines with equations $3y - 4x = 24$ and $3y + 4x = 24$, and the curve with equation $y = \frac{1}{9}x^2 + 3$. The place-holder is formed by rotating this cross-section about the $y$-axis, and is constructed from solid plastic.

Find the volume of plastic needed to construct the place-holder.

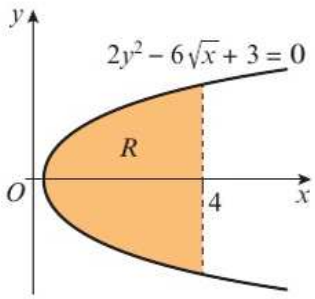



## Mixed exercise 5

E **1** The curve shown in the diagram has equation $y = x^2\sqrt{9 - x^2}$. The finite region $R$ is bounded by the curve and the $x$-axis. The region is rotated through $2\pi$ radians about the $x$-axis to generate a solid of revolution. Find the exact value of the volume of the solid that is generated. **(5 marks)**




P **2** The diagram shows the curve with equation $2y^2 - 6\sqrt{x} + 3 = 0$. The shaded region is bounded by the curve and the line $x = 4$.



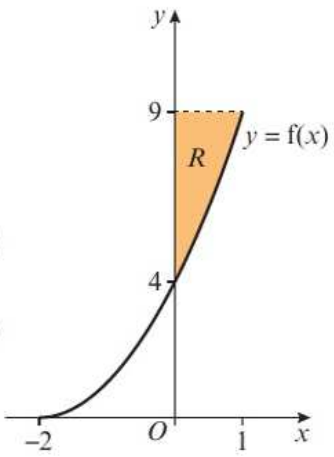

**a** Find the value of $x$ at the point where the curve cuts the $x$-axis.

The region is rotated about the $x$-axis to generate a solid of revolution.

**b** Find the volume of the solid generated.

E **3** $f(x) = x^2 + 4x + 4, x \geqslant -2$

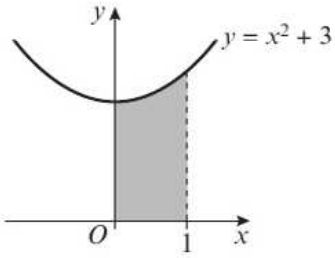
The diagram shows the finite region $R$ bounded by the curve $y = f(x)$, the $y$-axis and the lines $y = 4$ and $y = 9$.




**a** Show that the equation $y = f(x)$ can be written as $x^2 = 4 - 4\sqrt{y} + y$. **(2 marks)**

**b** The region $R$ is rotated through $2\pi$ radians about the $y$-axis. Find the exact volume of the solid generated. **(5 marks)**

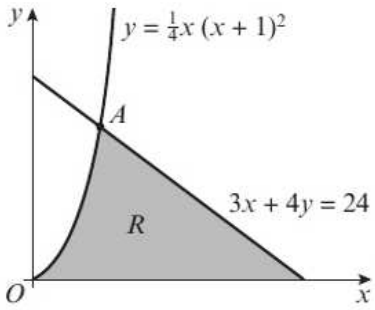
E **4** The diagram shows the shaded region bounded by the curve with equation $y = x^2 + 3$, the line $x = 1$, the $x$-axis and the $y$-axis. Find the volume generated when the region is rotated through $2\pi$ radians:




**a** about the $x$-axis **(3 marks)**

**b** about the $y$-axis. **(4 marks)**

E/P **5** The diagram shows the curve with equation $y = \frac{1}{4}x(x + 1)^2$ and the line with equation $3x + 4y = 24$. The line and the curve intersect at the point $A$.

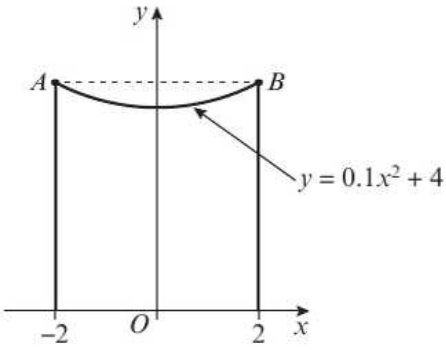



**a** Show that the coordinates of the point $A$ are (2, 4.5). **(2 marks)**

The shaded region $R$ is bounded by the curve, the line and the $x$-axis. The region is rotated through $2\pi$ radians about the $x$-axis.

**b** Find the exact volume of solid generated. **(6 marks)**

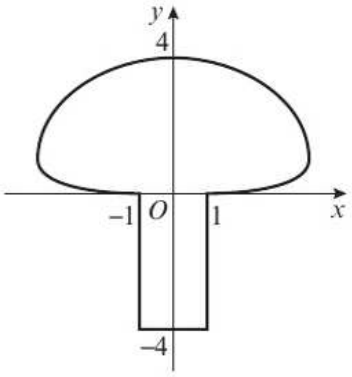
E/P **6** The diagram shows a cross-section of a circular golf ball trophy holder. The dimensions shown on the diagram are in centimetres. The cross-section of the trophy is formed by the lines $x = -2$, $x = 2$, the $x$-axis and the curve with equation $y = 0.1x^2 + 4$. The cross-section is rotated 360° about the $y$-axis. The trophy holder is to be cast out of solid bronze.




**a** Use this model to find the volume of bronze needed to make the trophy. **(5 marks)**

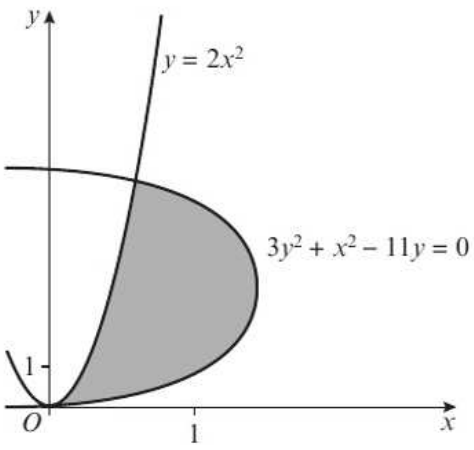
**b** Give one limitation of this model. **(1 mark)**

E/P **7** The diagram shows the outline of a circular mushroom. The dimensions on the diagram are in centimetres. The cap of the mushroom is modelled by the curve with equation $\frac{1}{4}x^2 - 8\sqrt{y} + 4y = 0$. The mushroom is formed by rotating the shape shown about the $y$-axis. Find the exact volume of the mushroom. **(6 marks)**



E/P **8** The shaded region is bounded by the curves with equations $y = 2x^2$ and $3y^2 + x^2 - 11y = 0$.

The shaded region is rotated 360° about the $y$-axis. Find the exact volume of the solid of revolution generated. **(9 marks)**

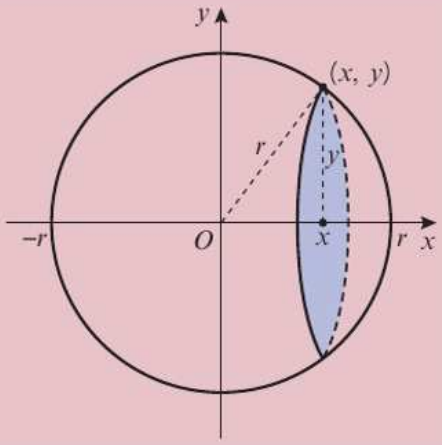



## Challenge

The diagram shows a sphere of radius $r$ and centre (0, 0).

**a** Show that the area of the shaded disc is $\pi(r^2 - x^2)$.

**b** By considering an integral over an appropriate interval, show that the volume of a sphere is $\frac{4}{3}\pi r^3$.




## Summary of key points

**1** The volume of revolution formed when $y = f(x)$ is rotated about the $x$-axis between $x = a$ and $x = b$ is given by

$$\textbf{Volume} = \pi \int_a^b y^2 \, dx$$

**2** The volume of revolution formed when $x = f(y)$ is rotated about the $y$-axis between $y = a$ and $y = b$ is given by

$$\textbf{Volume} = \pi \int_a^b x^2 \, dy$$

**3** A cylinder of height $h$ and radius $r$ has volume $\pi r^2 h$.

**4** A cone of height $h$ and base radius $r$ has volume $\frac{1}{3}\pi r^2 h$.